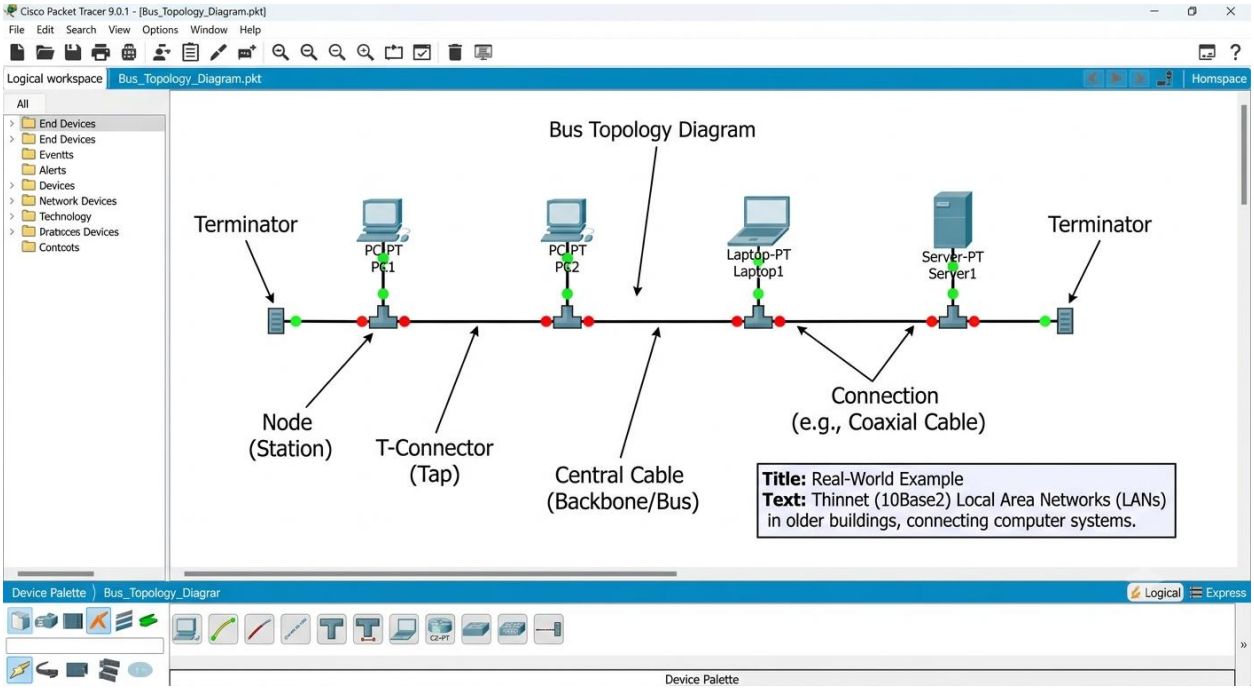
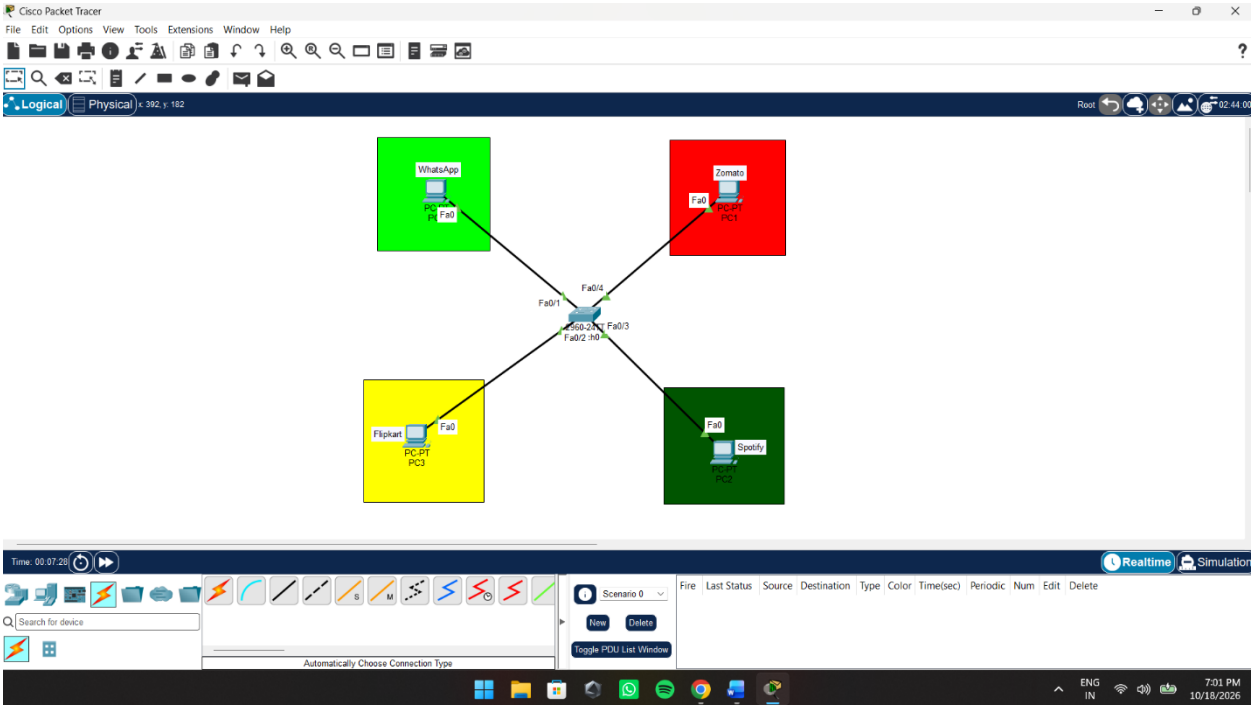
Assignment 18

Network Topologies Theory

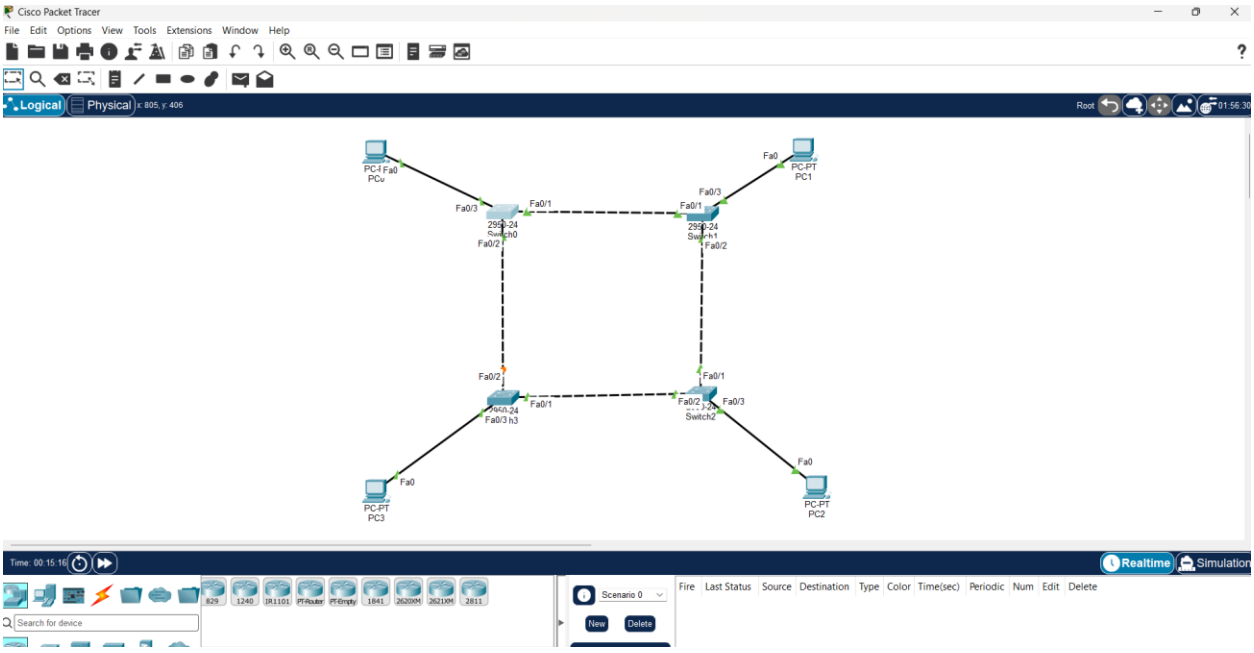
1. **Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.**



2. **In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.**



3. **Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup. Hint: Use arrows to show the direction of data flow.**



- In a ring topology, data travels in a single direction along a closed loop, passing through each device on the network. Each node acts as a repeater, boosting the signal and passing the information to the next device until it reaches its intended destination.

4. **Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.**

Topology	Advantages (2)	Disadvantages (2)
Mesh	1. Very reliable because multiple paths are available. 2. If one connection fails, data can use another path.	1. Expensive because it needs many cables. 2. Difficult to set up and maintain.
Star	1. Easy to set up and manage. 2. Easy to add or remove devices.	1. If the central switch fails, the whole network stops working. 2. Needs more cables than some other topologies.